

Bilinear Transformation: Type 1: Not  $\infty$  type  
Type 2:  $\infty$  type

The transformation

$$w = \frac{az + b}{cz + d}, \quad ad - bc \neq 0, \quad \text{where } a, b, c, d$$

are complex constant is called an

bilinear transformation.

$Z_t$  is also called - Möbius Transformation

(or) Linear fractional Transformation.

$$\frac{(1-j)-j}{j-1} \times \frac{1-z}{j-1} = \frac{(j-j)-0}{w-j-1} \times \frac{j-w}{0-j}$$

Note:

If  $w_1, w_2, w_3$  are the image of  
the pt  $z_1, z_2, z_3$  (respectively) under the  
bilinear transformation then

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bilinear transformation then

$$\frac{w_1 - w_2}{w_1 - w_3} \times \frac{w_2 - w_3}{w_3 - w} = \frac{z_1 - z_2}{z_1 - z_3} \times \frac{z_2 - z_3}{z_3 - z}$$

Componendo & Dividendo ratio

$$\frac{a}{b} = \frac{a+b}{a-b}$$

$$\frac{1b + 5b}{51b + b}$$

$$\frac{wb}{5b}$$

# Problem

1. Find the Bilinear Transformation which maps  $z = 1, i, -1$  on to  $w = i, 0, -i$

$$\text{Let } z_1 = 1; z_2 = i; z_3 = -1$$

$$w_1 = i; w_2 = 0; w_3 = -i$$

Let the Transformation be

$$\frac{w - w_1}{w_1 - w_2} \times \frac{w_2 - w_3}{w_3 - w} = \frac{z - z_1}{z_1 - z_2} \times \frac{z_2 - z_3}{z_3 - z}$$

$$\frac{w - i}{i - 0} \times \frac{0 - (-i)}{-i - w} = \frac{z - 1}{1 - i} \times \frac{i - (-1)}{-1 - z}$$

$$\frac{w - i}{i} \times \frac{i}{i + w} = \frac{z - 1}{1 - i} \times \frac{1 + i}{1 + z}$$

$$\frac{w - i}{i - 0} \times \frac{0 - (-i)}{-i - w} = \frac{z - 1}{1 - i} \times \frac{i - (-1)}{-1 - z}$$

$$\frac{w - i}{i} \times \frac{i}{i + w} = \frac{z - 1}{1 - i} \times \frac{1 + i}{1 + z}$$

$$\frac{w - i}{i + w} = \frac{(z - 1)(1 + i)}{(1 - i)(1 + z)}$$

$$\frac{w - i}{i + w} = \frac{z - 1 + iz - i}{1 - i + z - zi}$$

$$\frac{w - i + i + w}{w - i - i - w} = \frac{z - 1 + iz - i}{z - 1 + iz - i - 1 + i - z + z}$$

$$\frac{2w}{-2i} = \frac{2z - 2i}{-2 + 2iz}$$

$$\frac{w}{-i} = \frac{2(z-i)}{2(-1+iz)}$$

$$w = \frac{-2i(z-i)}{2(-1+iz)}$$

$$= \frac{-i(z-i)}{iz-1} \Rightarrow \frac{-iz+1}{iz-1}$$

$$w = \frac{-(1+iz)}{iz-1}$$

2) Find bilinear transformation  
 from  $z = -1, 0, 1$  onto  $w = 0, i, 3i$

$$\text{Let } z_1 = -1; z_2 = 0; z_3 = 1$$

$$w_1 = 0; w_2 = i; w_3 = 3i$$



2)  $z = -1, 0, 1$  Find bilinear transformation  
onto  $w = 0, i, 3i$

Let  $z_1 = -1$ ;  $z_2 = 0$ ;  $z_3 = 1$

$w_1 = 0$ ;  $w_2 = i$ ;  $w_3 = 3i$

$$\frac{w - w_1}{w_1 - w_2} \times \frac{w_2 - w_3}{w_3 - w} = \frac{z - z_1}{z_1 - z_2} \times \frac{z_2 - z_3}{z_3 - z}$$

$$\frac{w - 0}{0 - i} \times \frac{i - 3i}{3i - w} = \frac{z + 1}{-1 - 0} \times \frac{0 - 1}{1 - z}$$

$$\frac{w}{-i} \times \frac{-2i}{3i - w} = \frac{z + 1}{-1} \times \frac{-1}{1 - z}$$

$$\frac{2w}{3i - w} = \frac{z + 1}{1 - z}$$

$$2w(1 - z) = (z + 1)(3i - w)$$

$$2w - 2wz = 3i^0 z + 3i^0 - wz - w$$

$$2\check{w} - 2\check{w}z - 3i^0 z - 3i^0 + wz + \check{w} = 0$$

$$3w - wz = 3i^0 z + 3i^0$$

$$1 - w(3 - z) = 3i^0(z + 1)$$

$$w = \frac{3i^0(z+1)}{3-z}$$

③ Find Bilinear transformation which maps  $z = 1, i, -1$  on to  $w = 2, i, -2$

Let

$$z_1 = 1; z_2 = i; z_3 = -1$$

$$w_1 = 2; w_2 = i; w_3 = -2$$

$$w = \frac{-6z + 2i}{z - i} \quad \text{A}$$

$$\frac{w - w_1}{w_1 - w_2} \times \frac{w_2 - w_3}{w_3 - w} = \frac{z - z_1}{z_1 - z_2} \times \frac{z_2 - z_3}{z_3 - z}$$

③ Find Bilinear transformation which maps  $z = 1, i, -1$  on to  $w = 2, i, -2$

Let  $z_1 = 1; z_2 = i; z_3 = -1$

$w_1 = 2; w_2 = i; w_3 = -2$

$$\frac{w - w_1}{w_1 - w_2} \times \frac{w_2 - w_3}{w_3 - w} = \frac{z - z_1}{z_1 - z_2} \times \frac{z_2 - z_3}{z_3 - z}$$

$$\frac{w - 2}{2 - i} \times \frac{i + 2}{-2 - w} = \frac{z - 1}{1 - i} \times \frac{i + 1}{-1 - z}$$

$$\frac{w - 2}{2 - i} \times \frac{i + 2}{(2 + w)} = \frac{z - 1}{1 - i} \times \frac{i + 1}{(1 + z)}$$

$$\frac{wi + 2w - 2i - 4}{-4 - 2w + 2i + wi} = \frac{zi + z - i - 1}{-1 - z + i + zi}$$

$$\frac{wi + 2w - 2i - 4}{wi + 2w - 2i - 4 + 4 + 2w - 2i - wi} = \frac{zi + z - i - 1}{zi + z - i - 1 + 1 + z - i - z}$$



$$\frac{2w^0 - 8}{4w - 4i} = \frac{2z^0 - 2}{2z - 2i} \Rightarrow \frac{2(w^0 - 4)}{2(2w - 2i)} = \frac{2(z^0 - 1)}{2(z - i)}$$

$$(w^0 - 4)(z - i) = (z^0 - 1)(2w - 2i)$$

$$i^0 zw - i^2 w - 4z + 4i^0 = 2i^0 wz - 2w - 2i^2 z + 2i^0$$

$$i^0 zw + w - 4z + 4i^0 = 2i^0 wz - 2w + 2z + 2i^0$$

$$\underline{i^0 zw} + \underline{w} - 4z + 4i^0 - 2i^0 \underline{wz} + 2w - 2z - 2i^0 = 0$$

$$-i^0 wz + 3w - 6z + 2i^0 = 0$$

$$-w(i^0 z - 3) = 6z - 2i^0$$

$$w = \frac{-6z + 2i^0}{i^0 z - 3} //$$

Find Bilinear transformation which maps

$z = 0, -i, -1$  onto  $w = 0, i, \infty$

$$iz - 3 //$$

Find Bilinear transformation which maps  
 $z = 0, -i, -1$  onto  $w = 0, i, \infty$

$$\text{Let } z_1 = 0 ; z_2 = -i ; z_3 = -1$$

$$w_1 = 0 ; w_2 = i ; w_3 = \infty$$

$$\frac{w - w_1}{w_1 - w_2} \times \frac{w_2 - w_3}{w_3 - w} = \frac{z - z_1}{z_1 - z_2} \times \frac{z_2 - z_3}{z_3 - z}$$

$$\frac{w - w_1}{w_1 - w_2} \times \frac{w_3 \left( \frac{w_2}{w_3} - 1 \right)}{w_3 \left( 1 - \frac{w}{w_3} \right)} = \frac{z - z_1}{z_1 - z_2} \times \frac{z_2 - z_3}{z_3 - z}$$

$$\frac{w - 0}{0 - i} \times \frac{\left( \frac{i}{\infty} - 1 \right)}{\left( 1 - \frac{w}{\infty} \right)} = \frac{z - 0}{0 + i} \times \frac{-i + 1}{-1 - z}$$

$$\frac{w}{-i} \times \frac{(-1)}{1} = \frac{z(1-i)}{-i(1+z)}$$

$$w = \frac{-z(1-i)}{1+z}$$

⑤ Find Bilinear transformation which maps  $z = \infty, i, 0$  onto  $w = 0, i, \infty$

$$\text{Let } z_1 = \infty; z_2 = i; z_3 = 0$$

$$w_1 = 0; w_2 = i; w_3 = \infty$$

$$\frac{w-w_1}{w_1-w_2} \times \frac{w_2-w_3}{w_3-w} = \frac{z-z_1}{z_1-z_2} \times \frac{z_2-z_3}{z_3-z}$$

$$\frac{\omega - \omega_1}{\omega_1 - \omega_2} \times \frac{\omega_2 - \omega_3}{\omega_3 - \omega} = \frac{z - z_1}{z_1 - z_2} \times \frac{z_2 - z_3}{z_3 - z}$$

$$\frac{\omega - \omega_1}{\omega_1 - \omega_2} \times \frac{\omega_3 \left( \frac{\omega_2}{\omega_3} - 1 \right)}{\omega_3 \left( 1 - \frac{\omega}{\omega_3} \right)} = \frac{z_1 \left( \frac{z}{z_1} - 1 \right)}{z_1 \left( 1 - \frac{z_2}{z_1} \right)} \times \frac{z_2 - z_3}{z_3 - z}$$

$$\frac{(\omega - 0)}{(0 - i)} \times \frac{\left( \frac{i}{\infty} - 1 \right)}{\left( 1 - \frac{\omega}{\infty} \right)} = \frac{\left( \frac{z}{\infty} - 1 \right)}{\left( 1 - \frac{i}{\infty} \right)} \times \frac{i - 0}{0 - z}$$

$$\frac{\omega}{+i} \times \frac{(-1)}{1} = \frac{-1}{1} \times \frac{i}{+z}$$

$$\frac{\omega}{i} = \left( \frac{i}{z} \right) \times \frac{0 - \omega}{i - 0}$$

$$\omega = \frac{\left( \frac{i^2}{z} - 1 \right)}{z}$$

$$\omega = \frac{-1}{z} //$$